

36104 · ASSESSMENT 1

Reading Claims and Designing Alternatives

See what a chart claims, create an alternative view, and verify a machine's reading of its data.

Weight	30% of the subject · individual
Covers	Sessions 1-3
Artefacts	Three charts from the released pool (twelve entries across varied domains), from three different domains. Artefact 1 is the primary artefact used for the supervised critique.
Released	August 8
Supervised component	25-minute AI-restricted window at the start of the fourth live session (Week 5). Thursday 27 August 2026
Due	Friday 28 August 2026, 11:59 pm (Sydney time)
Submit	One ZIP: <code>A1_template.ipynb</code> · the <code>data/</code> folder · the three supplied chart images renamed <code>original_chart_1.png</code> to <code>original_chart_3.png</code>

The task

You will take **three sourced chart artefacts from the released pool, from three different domains**, and work them through the course pipeline: see what each claims, develop an alternative view with the Visual Vocabulary, and *verify* a machine's interpretation of the data. For **each artefact**, complete a full 9-field critique, an alternative view, a full Chart Choice Decision Record, and a claim audit. Artefact 1 is the **primary artefact**: the one whose critique is completed in the supervised, AI-restricted window. That supervised critique is the only requirement that differs across the three.

Part A. Seeing (the critiques)

For each artefact, complete the full 9-field critique: main claim; audience; visual task; directly visible; interpretation rather than observation; what is omitted; what misleads; what needs verifying; and a repaired caption. For artefact 1—the **primary artefact**—address all nine fields by hand in the supervised, AI-restricted window on 27 August, then transcribe and edit them for clarity without AI in `critique_1`. Complete `critique_2` and `critique_3` outside that window.

Critique field guide. *Main claim*: the one-sentence message the chart advances. *Audience*: the intended reader or user. *Visual task*: what the reader must do—for example compare, rank, locate or track change. *Directly visible*: what is literally encoded in marks, labels and scales. *Interpretation rather than observation*: what the design invites you to infer but does not directly show. *What is omitted*: relevant data, context, uncertainty or comparison left out. *What misleads*: how selection, encoding or framing—including title, caption, annotation or baseline—could create an unjustified impression. *What needs verifying*: the source, calculation or inference that requires checking. *Repaired caption*: a caption that states the supported finding and its important limit.

Part B. Choosing (the alternative views)

For each artefact, use the supplied source-data extract and state a claim that can be supported by that data. Create an alternative view designed to communicate that claim. Across the three artefacts, the alternatives must use valid Visual Vocabulary categories and represent **at least two different categories**. If you use the same category more than once, explain in each Decision Record why it is the best fit for that artefact's claim. Explain what each design makes easier and harder to see than

the published original and how those differences follow from the stated claim. Complete the full **Chart Choice Decision Record** for each artefact (`decision_record_1..3`).

Decision Record field guide. Identify the intended audience; the question they need answered; the decision the view supports; the Visual Vocabulary category; the comparison the reader must make; and the selected chart form. Then explain why that form fits, name one genuine alternative you considered, and give a specific reason for rejecting it.

Part C. Verifying (three claim audits)

Generate an interpretation of each dataset using the **supplied prompt** (verbatim, in the template). Classify **every claim** with the four-way taxonomy from the Seeing Data lab:

Label	Meaning
supported	the data directly backs it, attach an evidence probe
plausible but unverified	sounds right; this dataset cannot settle it
unsupported	asserted with no evidence in this data either way
contradicted	the data shows otherwise, attach an evidence probe

Probes are zero-argument functions returning the evidence (a filtered or aggregated DataFrame/Series): required for every **supported** and **contradicted** claim. Each audit must contain at least one claim that can be tested with a probe; do not manufacture other claim types simply to fill a quota. Finish with the five-question AI disclosure.

In the notebook, each claim is a short paraphrase used as the matching key in both the classification dictionary and, when required, the probe dictionary.

Your artefacts

Choose **three artefacts from the released pool, from three different domains**. Each of the twelve entries contains a published or source-linked chart, corresponding tidy source-data extract, data dictionary and provenance sheet. The approved printable chart-card examples are the only course-reproduction exception. Use the supplied data rather than digitising the image. Choose your primary before the supervised session and complete the data preparation, alternative view and other preparation in advance. On 27 August, record its pool ID on the supervised sheet. The pool ID entered for artefact 1 in the notebook—for example, `POOL-03`—must match the pool ID written on that sheet. Two students may study the same source chart, but every submission must be individual and independently produced.

Use the following canonical domain labels exactly in `META_n`:

Pool ID	Domain	Pool ID	Domain
POOL-01	climate science	POOL-07	economics
POOL-02	public health	POOL-08	ecology
POOL-03	astronomy	POOL-09	astronomy
POOL-04	wellbeing	POOL-10	energy
POOL-05	economics	POOL-11	geophysics
POOL-06	energy	POOL-12	transportation

Known source limitations. POOL-04's chart card shows ranks 1–48 while the supplied source contains all 143 ranked countries and does not contain the simultaneous rank intervals drawn in the chart. In POOL-06, a few published Sankey component labels do not add exactly to the displayed node totals; treat this as a documented source-chart limitation, not as a student calculation error. POOL-12 contains undirected historical airport-pair records: its two endpoints are alphabetically ordered, and `airline_record_count` is neither passenger nor flight volume. The individual provenance sheets repeat these details.

Background research and domain knowledge. The chart, source data and pool notes are starting points, not a complete package of everything you may need. Answering some questions well may require modest background reading about the topic, measures and context represented in the visualisation. You are not expected to become a domain expert, but you should learn enough to engage intelligently with the chart's claims, assumptions and limitations. You may use web search, library resources, reports and other credible sources for this purpose. If you use an AI system to locate or explain background information, independently verify its factual claims against credible sources. Keep a record of the sources that inform your analysis and cite them where relevant. You should also search for other visualisations of the same or closely related data to see how other designers have framed, filtered and encoded it and to develop ideas for your own alternative. Do not copy a design. If another visualisation materially influences your choices, identify it briefly in the relevant Decision Record—normally under `why_it_fits` or `reason_for_rejection`—and provide a working URL. A separate reference list is not required. Background research should support your own analysis of the chart; it does not replace close examination of the visualisation and its data.

Data-use rule. Copy the supplied CSV for each chosen chart into the submission's `data/` folder, rename it with its pool ID, and load that filename in `data_n`. Copy the publisher, publication value and source URL from the provenance sheet into the corresponding identifying-information block (`META_n`). Keep `data_n` as the unchanged DataFrame loaded from that exact file. You may create separately named filtered, reshaped or derived variables, but must preserve the source fields and document every transformation. Do not add values inferred from the image or present derived values as source measurements. Data handling and disclosure are judged under H2.

Reconstruction limit. The supplied extract may not reproduce every detail of the published original. Establish which components are supported by the supplied fields and which depend on unavailable data, transformations, modelling, annotations or design assets. Use this judgement when critiquing the original and explaining your alternative.

Copyright note. Include only the chart image needed for criticism and review, cite its publisher, creator (where known), date and source URL, and do not republish the assessment beyond the subject site. Follow any additional UTS library or copyright guidance supplied with the task.

AI policy

Part A's primary critique is drafted AI-free in the supervised window and may be edited for clarity without AI after transcription. An assistant may be used for Part B's coding and alternative-view work; you remain responsible for checking and accepting or rejecting its output. Part C requires three assistant-generated interpretations. Report your use or non-use of AI accurately. False or misleading disclosure is an academic-integrity issue.

Submission requirements

The template notebook contains a set of required named variables and functions for each artefact. These include the identifying-information blocks (`META_1`, `META_2`, and `META_3`), critiques (`critique_1`

to `critique_3`), datasets (`data_1` to `data_3`), alternative views (`redesign_1` to `redesign_3`), Decision Records, claim audits, evidence probes, and the AI disclosure. **Do not rename these items**, because the autograder checks them directly. The autograder executes your notebook top-to-bottom on a clean machine. **Run the self-check cell before submitting**: it runs the marker's G1–G6 checks (54 marks). G0's remaining 6 marks are awarded only after a clean top-to-bottom execution on the marking system. An execution error loses the G0 marks and may also prevent checks that depend on the affected code; it does not automatically erase otherwise verifiable marks. If an execution failure prevents fair assessment, the marker may allow one technical resubmission within 48 hours of notification. Only changes needed to restore execution are permitted; the analytical content must remain unchanged.

Rubric, 100 marks (worth 30% of the subject)

The rubric maximum is the sum of the criterion maxima: 60 automated marks plus 40 human-marked marks equals 100. Canvas records both the assignment and rubric out of 100. The Assignment 1 group contributes 30% of the subject, so no manual score conversion is required.

Automated (60 marks, scored by `a1_autograder.py`)

ID	Item	Marks	Check
G0	Notebook executes end-to-end	6	all cells run clean, no errors
G1	Three artefacts: identity and files valid	6	<code>STUDENT_ID</code> and all identifying-information blocks complete; three distinct IDs from <code>P00L-01..12</code> ; the corresponding canonical domains are distinct; URLs and <code>data_status</code> valid; each supplied original chart matches its selected pool item
G2	Data provenance ×3	9	each exact supplied <code>data/P00L-XX_source_data.csv</code> loads unchanged into <code>data_n</code> ; ≥ 6 rows $\times$ ≥ 2 columns; no empty columns; source identifying information and transformations documented
G3	Alternative-view mechanics ×3	15	each alternative view (<code>redesign_n</code>) returns its Axes; all categories are valid and at least two different Vocabulary categories are represented across the three artefacts; bar-family charts include a zero baseline; every chart title has at least 15 characters
G4	Claim-audit structure ×3	14	for each artefact: a 150–250-word interpretation from the supplied prompt; ≥ 4 validly labelled claims; at least one <code>supported</code> or <code>contradicted</code> claim; an executable, non-empty probe for every such claim
G5	Critiques + Decision Records ×3	6	all three 9-field critiques and all three 9-field Decision Records are complete; each <code>vocabulary_category</code> matches <code>CATEGORY_n</code>
G6	Five-question AI disclosure	4	all 5 answers completed; each answer has at least 3 words and is not a placeholder
Total		60	

Human-marked (40 marks, anchored 0/1/2 per item)

Item (weight)	0	1	2
H1 · Critique insight across three (×5)	Restates the charts	Correctly identifies claims and real omissions	Also separates <i>directly visible</i> from <i>interpretation</i> , explains how title, caption, annotation or baseline frames the claim where relevant, and the repaired captions actually repair
H2 · Data handling, across artefacts (×2.5)	Source data misread or transformations undisclosed	Units and transformations mostly correct	Source fields preserved; transformations correct, reproducible and disclosed
H3 · Decision Records across three (×5)	No argument, or category labels only	Correct task categories with generic justifications	Ties audience → task → channel precision; rejected alternatives are real and reasons specific
H4 · Alternative-view craft, across the three (×5)	Unreadable or misleading	Honest defaults, readable	Titles state findings; scales honest; labels carry the argument
H5 · Audit judgement across three (×2.5)	Classifications mostly wrong	Labels mostly appropriate, but probes are weak	Labels distinguish evidence from plausibility precisely; every required probe directly tests the claim
Total: 40 marks (weights × anchor score; maximum $2 \times (5 + 2.5 + 5 + 5 + 2.5) = 40$)			

Process requirements (not marked, but gates)

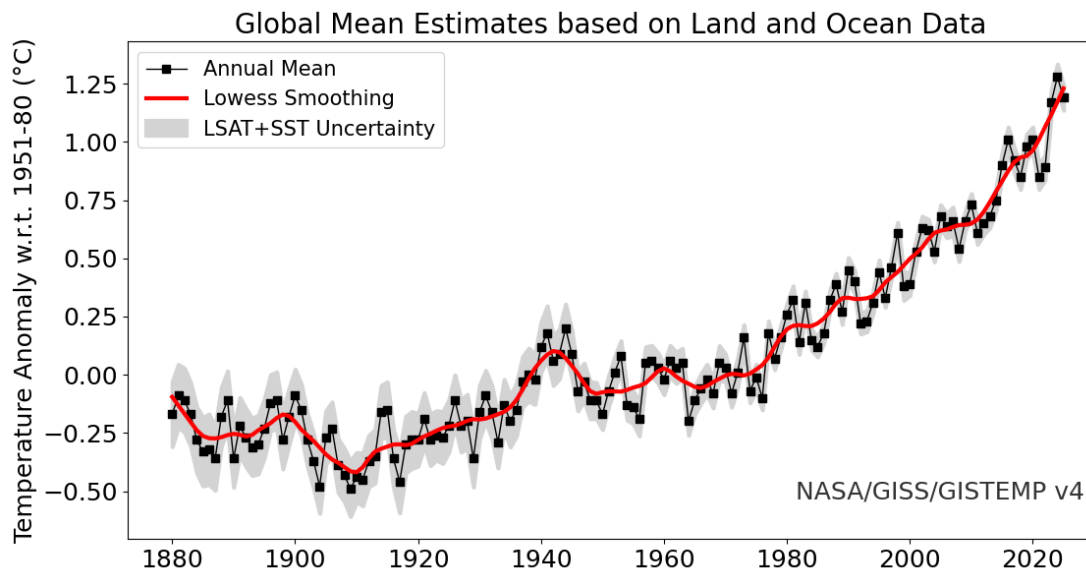
- Part A supervised notes handed in during the 27 August window. Missing notes → H1 capped at 0.
- The pool ID entered for artefact 1 in the notebook must match the pool ID written on the supervised sheet (27 August).

Artefact pool: charts and sources

The twelve chart images are reproduced below for assignment reference. The downloadable pool supplies the corresponding chart image, CSV, data dictionary and provenance sheet for each entry.

POOL-01 · Deviation

Domain: climate science · Global mean temperature anomaly · NASA Goddard Institute for Space Studies



Source: [Open the chart and source page](#)

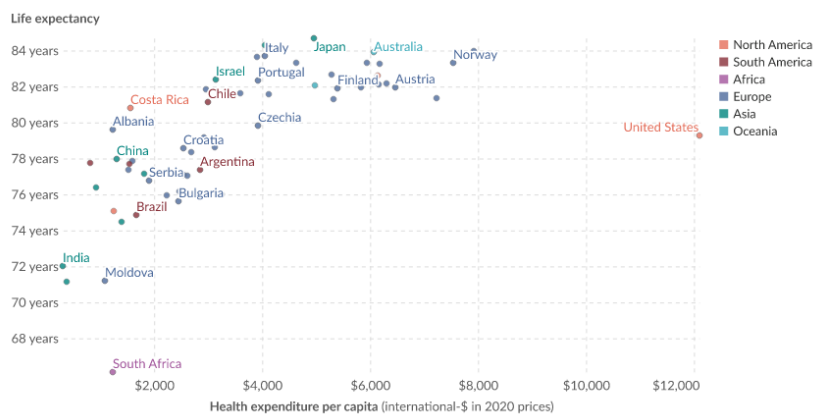
Data: [Download the publisher's source data](#)

POOL-02 · Correlation

Domain: public health · Life expectancy vs health expenditure · Our World in Data

Life expectancy vs. health spending 2023

The period life expectancy at birth, in a given year. Health expenditure includes all financing schemes and covers all aspects of healthcare. This data is adjusted for inflation and differences in living costs between countries.



Data source: UN, World Population Prospects (2024); OECD (2026)

Note: Health expenditure is expressed in international-\$ at 2020 prices.

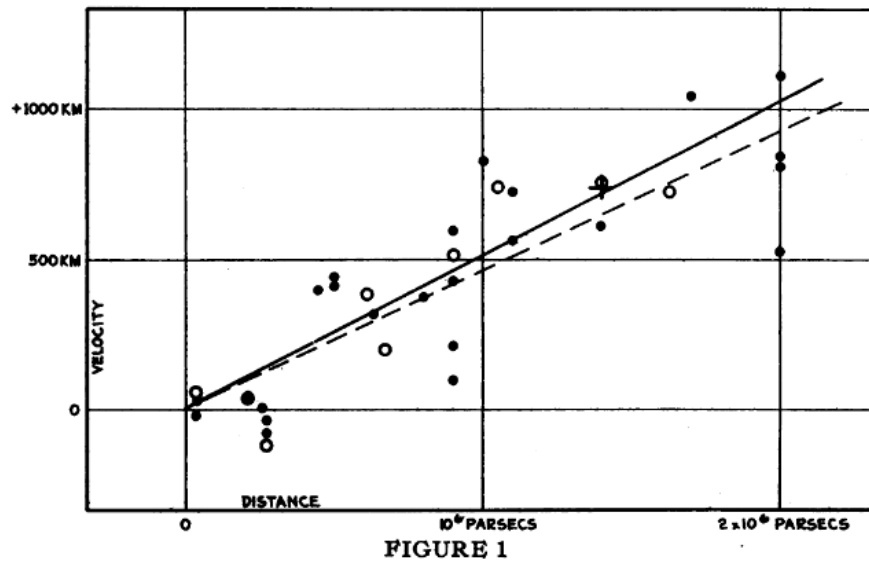
OurWorldinData.org/financing-healthcare | CC BY

Source: [Open the chart and source page](#)

Data: [Download the publisher's source data](#)

POOL-03 · Correlation

Domain: astronomy · Hubble's 1929 distance–velocity relation · Proceedings of the National Academy of Sciences

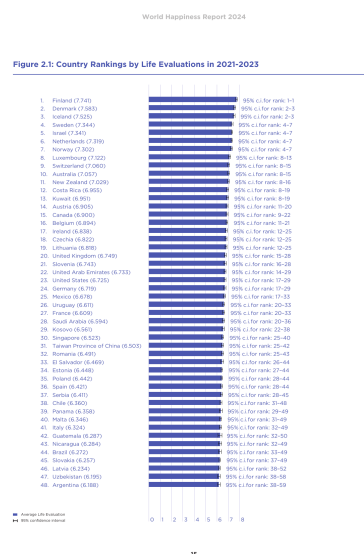


Source: [Open the chart and source page](#)

Data: [Open the published tabulated observations](#)

POOL-04 · Ranking

Domain: wellbeing · Country rankings by life evaluation, 2021–2023 · World Happiness Report

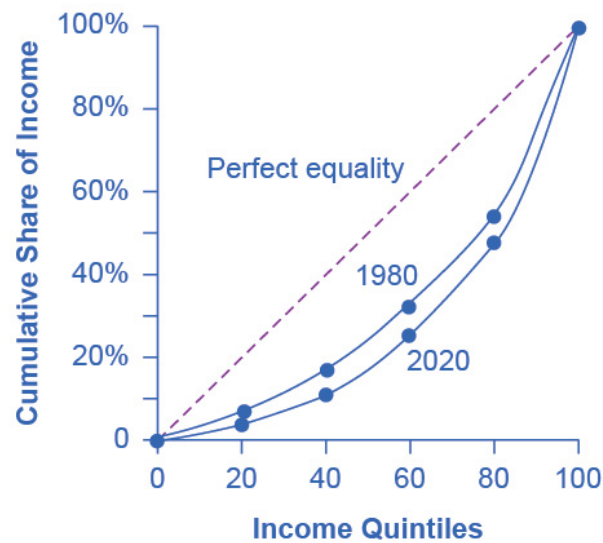


Source: [Open the chart and source page](#)

Data: [Download the report's supporting data](#)

POOL-05 · Distribution

Domain: economics · Lorenz curves for US family income · OpenStax

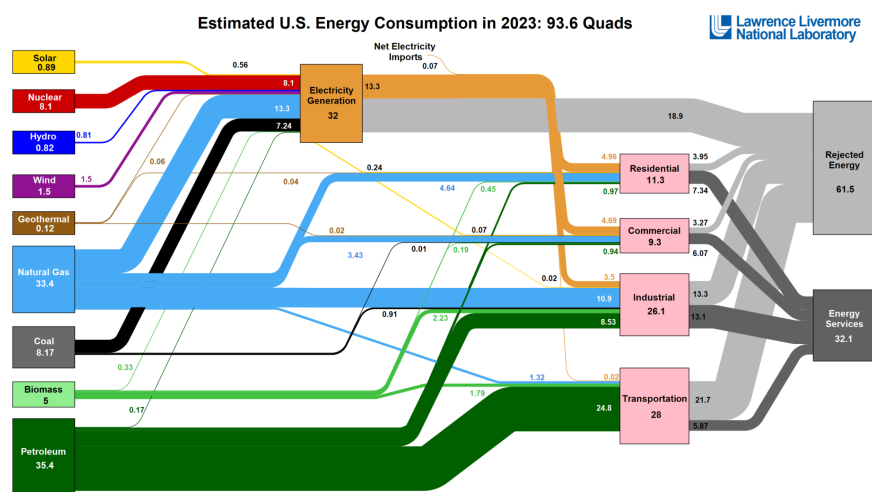


Source: [Open the chart and source page](#)

Data: [Open the publisher's chart and source table](#)

POOL-06 · Flow

Domain: energy · Estimated US energy consumption, 2023 · Lawrence Livermore National Laboratory / US Department of Energy

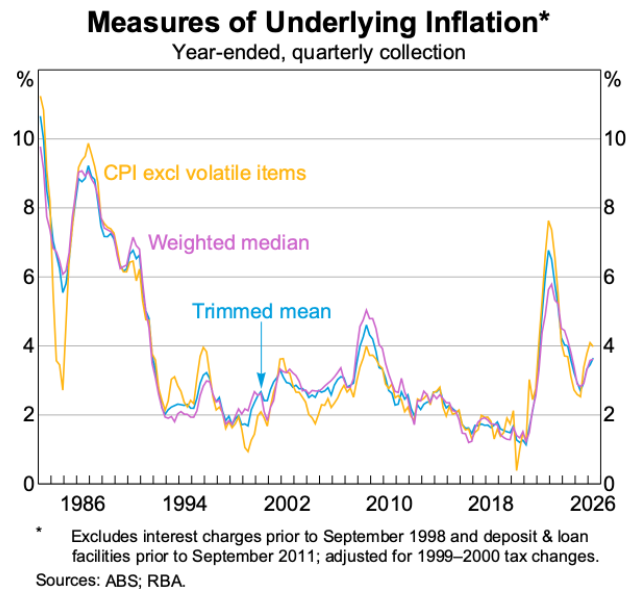


Source: [Open the chart and source page](#)

Data: [Open the published diagram; flow values are printed on the chart](#)

POOL-07 · Change over time

Domain: economics · Measures of underlying inflation · Reserve Bank of Australia

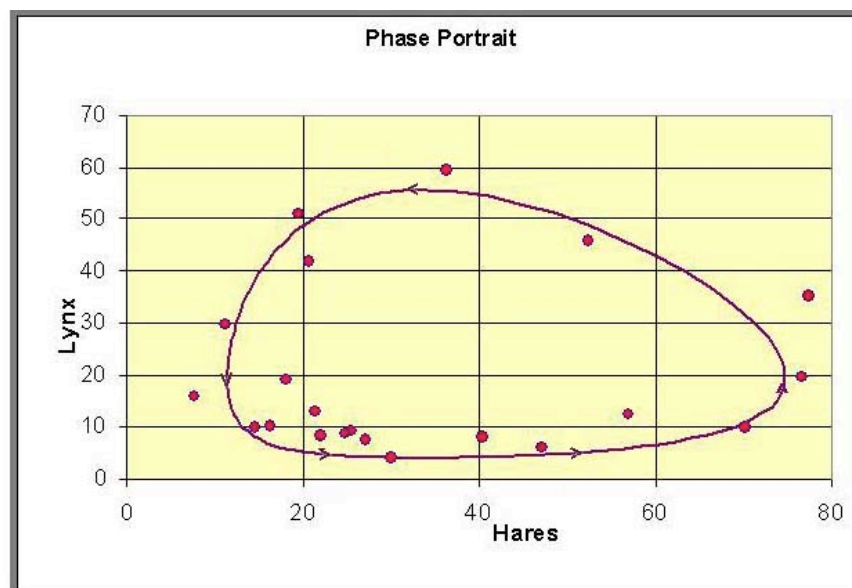


Source: [Open the chart and source page](#)

Data: [Download RBA table G1 data](#)

POOL-08 · Correlation

Domain: ecology · Lynx–hare phase portrait · San Diego State University (J. M. Mahaffy)

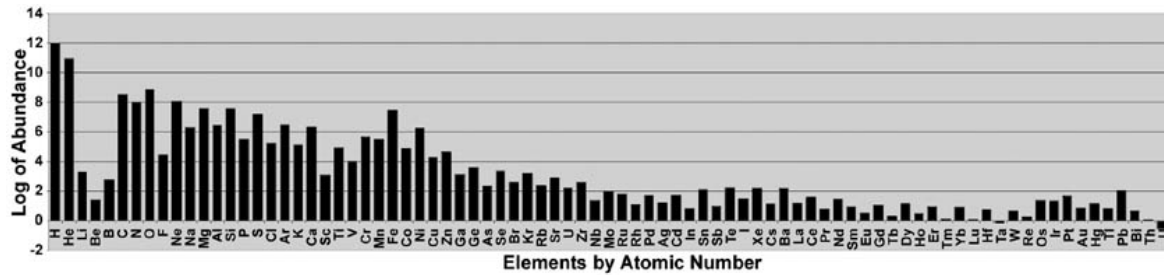


Source: [Open the chart and source page](#)

Data: [Open the published observations](#)

POOL-09 · Magnitude

Domain: astronomy · Cosmic abundance of the elements · NASA Imagine the Universe / HEASARC

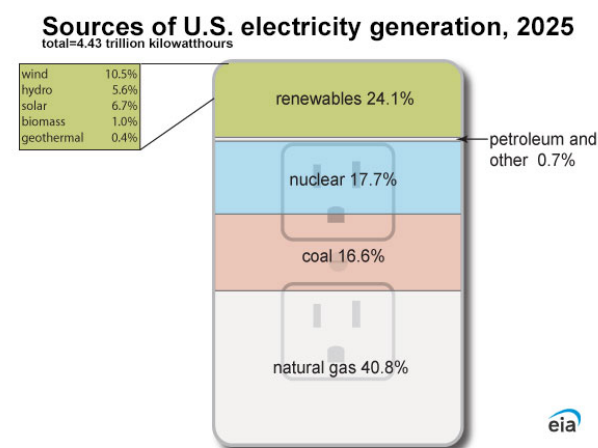


Source: [Open the chart and source page](#)

Data: [Open the referenced abundance table](#)

POOL-10 · Part-to-whole

Domain: energy · Sources of US electricity generation · US Energy Information Administration



Data source: U.S. Energy Information Administration, *Electric Power Monthly*, February 2026, preliminary data

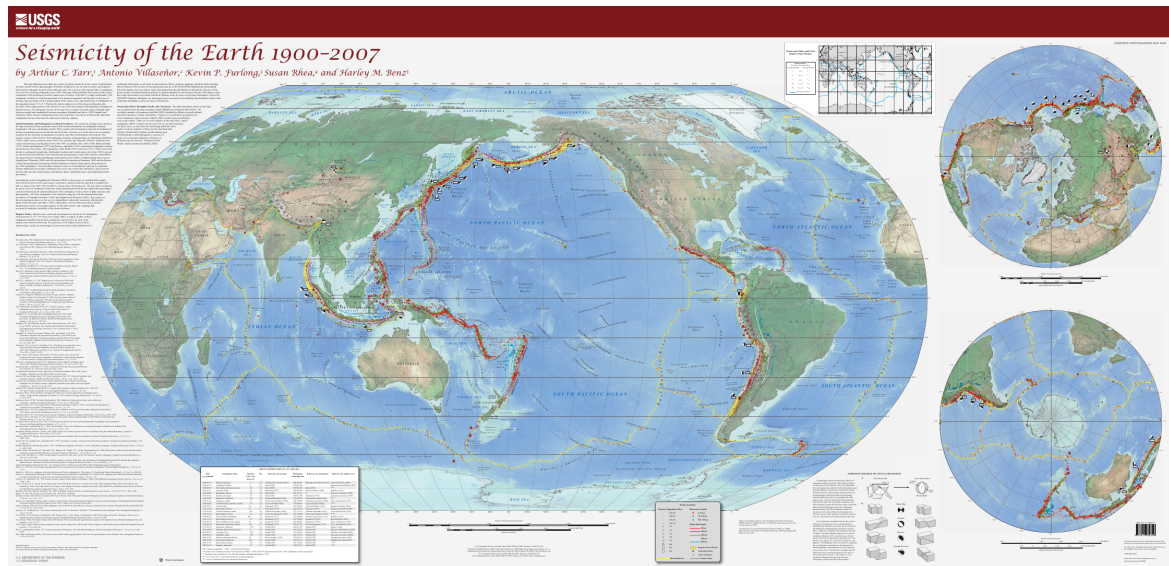
Note: Includes electricity generation from power plants with at least 1,000 kilowatts of electric generation capacity (utility-scale). Hydro is conventional hydroelectric. Petroleum includes petroleum liquids, petroleum coke, other gases (blast furnace gas and other manufactured and waste gases derived from fossil fuels), hydroelectric pumped storage, and other sources (non-biogenic municipal solid waste, batteries, hydrogen, purchased steam, sulfur, tire-derived fuel, and other miscellaneous energy sources). Sum of percentages may not equal 100% because of independent rounding.

Source: [Open the chart and source page](#)

Data: [Open the publisher's chart and source values](#)

POOL-11 · Spatial

Domain: geophysics · Seismicity of the Earth 1900–2007 · US Geological Survey

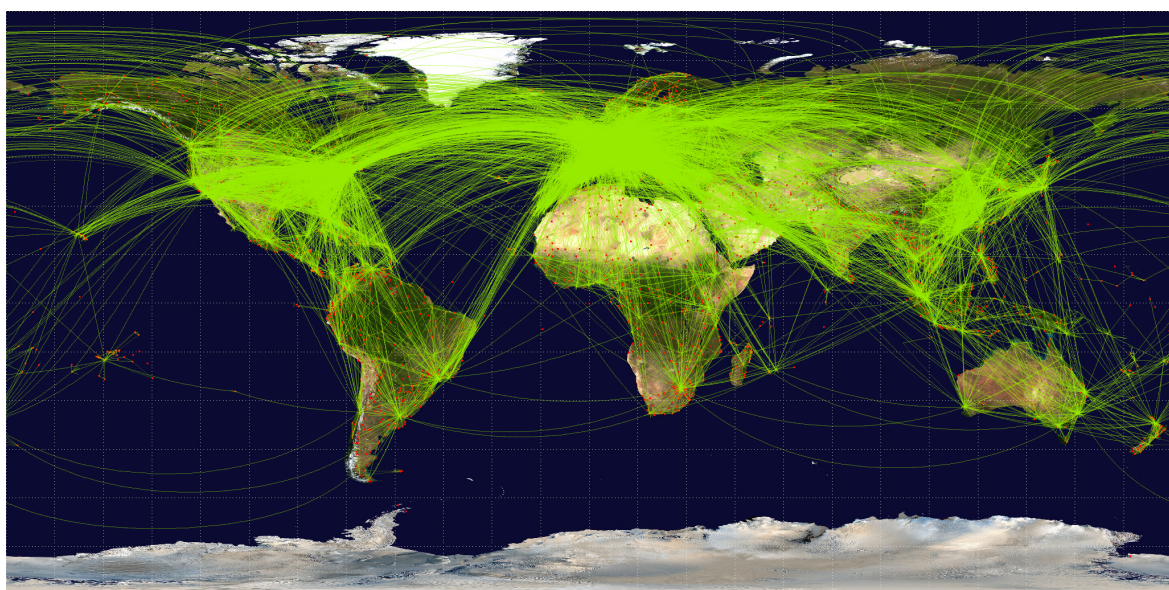


Source: [Open the chart and source page](#)

Data: [Download the documented USGS catalogue extract \(1900–2007, \$M \geq 7.3\$ \)](#)

POOL-12 · Flow

Domain: transportation · Worldwide airline route network · OpenFlights



Source: [Open the chart and source page](#)

Data: [Download the OpenFlights route data](#)